Topology
Home Work 5

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Exercise 1. Let $(X,\mathcal{U})$ be a topological space.

(a) Let $C \subseteq X$ be connected. Prove that its closure $\overline{C}$ is connected.

(b) Let $C_1, C_2 \subseteq X$ be connected and assume that $C_1 \cap C_2 \neq \emptyset$. Prove that $C_1 \cup C_2$ is connected.

(c) Let $p \in X$, and consider the connected component

$$C(p) := \bigcup_{\substack{C \subseteq X \text{ connected} \\ p \in C}} C \tag{1.1}$$

Prove that $C(p)$ is connected and closed in X.

Answer:

(1)(a) $(X,\mathcal{U})$ a topological space. We are given that $C \subset X$ is connected.
Let $\overline{C} \supset C$ be the closure of C and assume that it is not connected. Then $\exists\, U, V \in \mathcal{U}_{\overline{C}}$ both nonempty and disjoint such that:

$$U \cup V = \overline{C}.$$

As C is connected it is clear that C itself is contained in U or V (say U). If this were not the case we would have found two open disjoint sets (i.e. $U \cap C$ and $V \cap C$) in $\mathcal{U}_C$ that seperated C. Consequently, we must have that:

$$C \subset U \quad \text{and} \quad \overline{C} \cap V \neq \emptyset.$$

As $\overline{C} \cap V \neq \emptyset$ let $p \in \overline{C} \cap V$, i.e. $p \in \overline{C} \setminus C$ (i.e. an accumulation point of C). But, by definition, if $p \in \overline{C} \setminus C$

$$\Rightarrow \quad \forall\, O \in \mathcal{U} \text{ s.t. } p \in O \quad \Longrightarrow \quad O \cap C \neq \emptyset.$$

But V is one such $O \cap \overline{C}$ and hence:

$$\emptyset \neq V \cap C \subset V \cap U$$

<u>contradiction</u>.

Hence C connected $\Rightarrow \overline{C}$ also connected.

□

(b) Assume $C_1, C_2 \subset X$ are both connected. Given $C_1 \cap C_2 \neq \emptyset$ show $C_1 \cup C_2$ connected.

Assume $C_1 \cup C_2$ is not connected. Then $\exists\, U, V \in \mathcal{U}_{C_1 \cup C_2}$ s.t. U, V are disjoint and non-empty, and

$$U \cup V = C_1 \cup C_2.$$

Clearly C_1 & C_2 – both connected – must reside in different (U, V). Say $C_1 \subset U$ & $C_2 \subset V$ - use same argument as in part (a) to show this.

But

$$C_1 \cap C_2 \subset U \cap V$$
$$\text{and} \quad C_1 \cap C_2 \neq \emptyset \Rightarrow U \cap V \neq \emptyset.$$

Contradiction

Hence $C_1 \cup C_2$ is connected.

□

(c) Let

$$C(p) = \bigcup_{\substack{p \in C \\ C \text{ connected}}} C.$$

Assume $C(p)$ is not connected. Then $\exists\, U, V \in \mathcal{U}_{C(p)}$ both open, disjoint, non-empty and $C(p) = U \cup V$.
A similar argument to part (b) follows. As each C in $C(p)$ is connected it will belong to either U or V. U and V are both nonempty by assumption. This means that both U and V will contain different connected C's. Consequently, we must have that: $p \in U$ and $p \in V \implies p \in U \cap V$.

contradiction.

Hence $C(p)$ is connected.

Now $C(p)$ connected $\Rightarrow \overline{C(p)}$ connected (in part (1)).

$\Rightarrow C(p) \subset \overline{C(p)} \subset C(p)$ from the definition of $C(p)$ (1.1).
$\Rightarrow C(p) = \overline{C(p)}$
$\Rightarrow C(p)$ is closed.

□

Exercise 2.

A topological space $(X,\mathcal{U})$ is called T_1 if for every pair of distinct points $x, y \in X$, there exists an open set U such that

$$x \in U, \qquad y \notin U,$$

and an open set V such that

$$y \in V, \qquad x \notin V.$$

Notice that the open sets U and V are not required to be disjoint.

(a) Prove that a topological space $(X,\mathcal{U})$ is T_1 if and only if every singleton $\{x\}$ is closed in X.

Consider on $\mathbb{R}$ the topology of right rays

$$\mathcal{T} = \{(a, +\infty) : a \in \mathbb{R}\} \cup \{\emptyset, \mathbb{R}\}.$$

(b) Is $(\mathbb{R}, \mathcal{T})$ Hausdorff? Is $(\mathbb{R}, \mathcal{T})$ a T_1 space? Justify your answers.

Answer:

(a) Let $(X,\mathcal{U})$ be a topological space.
(i) Assume every singleton set $\{x\}$ $(x \in X)$ is closed in X.
Given x, y $(x \neq y)$ in X. Then:

$$\begin{aligned} U = X \setminus \{x\} \quad &\text{is open i.e. } U \in \mathcal{U} \\ V = X \setminus \{y\} \quad &\text{is open i.e. } V \in \mathcal{U}. \end{aligned}$$

Clearly:

$$\begin{aligned} \qquad\quad y \in U \quad &\text{and} \quad x \notin U. \\ \& \quad x \in V \quad &\text{and} \quad y \notin V. \end{aligned}$$

Hence $(X,\mathcal{U})$ is T_1.

(ii) Assume $(X,\mathcal{U})$ is T_1.
Given $x \in X$, then $\forall\, y \in X \setminus \{x\}$ - since $(X,\mathcal{U})$ is T1 - there is an open set U_y such that: $y \in U_y$ & $x \notin U_y$.

Define:

$$U = \bigcup_{y \in X \setminus \{x\}} U_y.$$

Then U is also open (i.e. $U \in \mathcal{U}$).
Then $X \setminus U = \{x\}$ is closed.

□

(b)

$$\mathcal{T} = \{(a, +\infty) \mid a \in \mathbb{R}\} \cup \{\emptyset, \mathbb{R}\}$$

is a topology on $\mathbb{R}$.

To determine if $\mathcal{T}$ is Hausdorff we must check that if given any $x, y \in \mathbb{R}$ $(x < y)$ there are disjoint open sets that contain x and y.
All open sets that contain x are of the form $(p, +\infty)$ where $p < x$ - or $\mathbb{R}$ itself.
All open sets containing y are of the form $(q, +\infty)$ where $q < y$ - or $\mathbb{R}$ itself.
As $x < y$. Any open set containing x must also contain $y \Rightarrow (\mathbb{R}, \mathcal{T})$ is not Hausdorff.

The same argument shows that any open set that contains x must contain y. So $(\mathbb{R}, \mathcal{T})$ is also not T_1.

Exercise 3.
A map $f : X \to Y$ between topological spaces is called open if, for every open set $U \subseteq X$, $f(U)$ is open in Y.

(a) Given $(X, \mathcal{U})$ and $(Y, \mathcal{V})$ topological spaces, show that, if $f : X \to Y$ is bijective, continuous and open, then f is a homeomorphism.

Consider the torus $T \subseteq \mathbb{R}^3$, endowed with the topology $\mathcal{U}_T$ induced by the Euclidean topology on $\mathbb{R}^3$. Consider also $S^1 \times S^1$ with the product topology.
Recall from class the following map, given in angular coordinates,

$$\begin{aligned} F : S^1 \times S^1 &\longrightarrow T, \\ (\theta, \varphi) &\longmapsto \big((2 + \cos(2\pi\theta)) \cos(2\pi\varphi), (2 + \cos(2\pi\theta)) \sin(2\pi\varphi), \sin(2\pi\theta)\big). \end{aligned}$$

(b) Prove that F is a homeomorphism.

Exercise 4. Let $(X,\mathcal{U})$ and $(Y,\mathcal{V})$ be non-empty topological spaces and consider $X \times Y$ endowed with the product topology. Prove that $X \times Y$ is connected if and only if both X and Y are connected.

Answer:

(i) Given that $(X \times Y,\mathcal{U}_{X\times Y})$ is connected prove that $(X,\mathcal{U}_X)$ and $(Y,\mathcal{U}_Y)$ are both connected.

First we construct a map:

$$f : (X \times Y,\mathcal{U}_{X\times Y}) \to (X,\mathcal{U}_X)$$
$$f(x,y) = x$$

This map is continuous since, for $O \in \mathcal{U}_X \implies f^{-1}(O) = O \times Y$ - which is open in $\mathcal{U}_{X\times Y}$. This map is also clearly surjective. So, given that $(X \times Y,\mathcal{U}_{X\times Y})$ is connected we must have that $(X,\mathcal{U}_X)$ is also connected.

The same argument can be applied to show that $(Y,\mathcal{U}_Y)$ must also be connected.

(ii) Given that $(X,\mathcal{U})$ and $(Y,\mathcal{V})$ are both connected.

Suppose $X \times Y = U \cup V$ is a separation into non-empty disjoint open sets. Choose $(x_0, y_0) \in U$. The connected slice $\{x_0\} \times Y$ lies entirely in U. For every $y \in Y$, the connected slice $X \times \{y\}$ intersects it at (x_0, y), so also lies entirely in U.

Thus $U = X \times Y$, contradicting $V \neq \varnothing$.

Exercise 5. Let $(X,\mathcal{U})$ and $(Y,\mathcal{V})$ be non-empty topological spaces and consider $X \times Y$ endowed with the product topology $\mathcal{U}_{X\times Y}$. Prove that $(X \times Y,\mathcal{U}_{X\times Y})$ is Hausdorff if and only if both $(X,\mathcal{U})$ and $(Y,\mathcal{V})$ are Hausdorff.

Answer:

(a) Assume $(X,\mathcal{U}_x),(Y,\mathcal{U}_y)$ are both Hausdorff. Let (x_1,y_1) & (x_2,y_2) be two distinct points in $X \times Y$.
(i) Assume $x_1 \neq x_2$ & $y_1 \neq y_2$. Then

$$\begin{aligned} & \exists\, U_1, U_2 \in \mathcal{U}_x \text{ s.t. } x_1 \in U_1,\ x_2 \in U_2, \quad U_1 \cap U_2 = \emptyset \\ \& \quad & \exists\, V_1, V_2 \in \mathcal{U}_y \text{ s.t. } y_1 \in V_1,\ y_2 \in V_2, \quad V_1 \cap V_2 = \emptyset. \end{aligned}$$

Then

$$\begin{aligned} & (x_1,y_1) \in U_1 \times V_1; \quad (x_2,y_2) \in U_2 \times V_2 \\ \& \quad & (U_1 \times V_1) \cap (U_2 \times V_2) = \emptyset. \end{aligned}$$

Hence: $(X \times Y,\mathcal{U}_{X\times Y})$ is Hausdorff.

(ii) If $y_1 = y_2 = y$ we just take $V \in \mathcal{U}_y$ s.t. $y \in V$. Again

$$\begin{aligned} & (x_1,y) \in U_1 \times V \quad \& \quad (x_2,y) \in U_2 \times V \\ \& \quad & (U_1 \times V) \cap (U_2 \times V) = \emptyset. \end{aligned}$$

Hence: $(X \times Y,\mathcal{U}_{X\times Y})$ is again Hausdorff.

(Same logic if $x_1 = x_2$ & $y_1 \neq y_2$).

$$\therefore \quad (X \times Y,\mathcal{U}_{X\times Y}) \text{ is Hausdorff}$$

(b) Assume that $(X \times Y,\mathcal{U}_{X\times Y})$ is Hausdorff.
Take any two distinct points $(x_1,y),(x_2,y) \in X \times Y$. As $X \times Y$ Hausdorff there must exist two open and disjoint "rectangles" R_1 & R_2 s.t. $(x_i,y) \in R_i$.

Let $R_1 = U_1 \times V_1$ and $R_2 = U_2 \times V_2$ - where $U_i \in \mathcal{U}_X$ and $V_i \in \mathcal{U}_Y$.
Let $V = V_1 \cap V_2$, then $U_1 \times V$ and $U_2 \times V$ are also open disjoint "rectangles" in $X \times Y$ that seperate the two points (x_1,y) and (x_2,y) - and their being disjoint means that $U_1 \cap U_2 = \emptyset$.
We conclude that $(X,\mathcal{U}_X)$ is Hausdorf.

The exact same argument can be applied for $(Y,\mathcal{U}_Y)$ to show it is also Hausdorff.

Hence we must have that both $(X,\mathcal{U}_X)$ and $(Y,\mathcal{U}_Y)$ are Hausdorff. □

Exercise 6. *(Universal property of the Product Topology)*
Let $(X_\alpha, \mathcal{U}_\alpha)_{\alpha \in A}$ be a family of topological spaces. Consider the set $X := \prod_{\alpha \in A} X_\alpha$ and, for every $\alpha \in A$, the projection $p_\alpha : X \to X_\alpha$. Prove that the product topology $\mathcal{U}_X$ on X is the unique topology of X such that, for every topological space $(Y, \mathcal{V})$ and map $f : Y \to X$, f is continuous if and only if $p_\alpha \circ f : Y \to X_\alpha$ is continuous for every $\alpha \in A$.

$$\begin{array}{ccc} (Y, \mathcal{V}) & \xrightarrow{f} & (X, \mathcal{U}_X) \\ & \searrow^{p_\alpha \circ f} & \downarrow p_\alpha \\ & & (X_\alpha, \mathcal{U}_\alpha) \end{array}$$

Answer: